

Programme	B. Sc. Statistics				
Course Code	STA5EJ303				
Course Title	Biostatistics				
Type of Course	Major Elective				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course Summary Objective	The student will describe the need and ethics of clinical trials and designs for various phases of clinical trials.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	The student will explain Principles of Biostatistical study designs	U	C	Instructor-created exams / Quiz
CO2	The student will explain measures of morbidity.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	The student will describe the concepts of survival time functions of important parametric models.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	The student will explain types of censoring and estimation of parameters using censored data.	U	C	Instructor-created exams / Home Assignments
CO5	The student will explain the non-parametric methods for estimating survival function and variance of the estimator using Kaplan –Meier methods.	Ap	P	One Minute Reflection Writing assignments

CO6	The student will describe the basic biological concepts in genetics.	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Introduction		14	20
	1	Examples of statistical problems in Biomedical Research		
	2	Types of Biological data		
	3	Principles of Biostatistical design of medical studies		
	4	study designs- observational study, experimental study-comparative experiment, cross over experiment		
	5	prospective and retrospective study		
	6	case-control and longitudinal study		
	7	Measuring the occurrence of disease, Measures of morbidity - prevalence and incidence rate, association between prevalence and incidence, uses of prevalence and incidence.		
	Sections from References:			
II	Survival analysis		12	20
	8	Introduction to survival analysis, concepts and definitions		
	9	Survival function		
	10	probability density function		
	11	hazard function		
	12	inter relationships,		
	13	Survival distributions- exponential distribution, Weibull distribution and lognormal distribution.		
	Sections from References:			
III	Types of censoring		10	15

	14	Concepts of censoring and truncation		
	15	Type I, Type II and progressive or random censoring with biological examples,		
	16	Estimation of mean survival time and variance of the estimator for type I and type II censored data with numerical examples (for exponential distribution).		
	17	Non-parametric methods for estimating survival function and variance of the estimator- Kaplan –Meier methods.		
	Sections from References:			
IV			12	15
	18	Basic biological concepts in genetics Mendel’s law, Hardy- Weinberg equilibrium		
	19	Random mating, natural selection, mutation, genetic drift,		
	20	Detection and estimation of linkage in heredity		
	21	Planning and design of clinical trials, Phase I, II, and III trials.		
	22	Ethics behind randomized studies involving human subjects; randomized dose-response studies (concept only)		
	Sections from References:			
V	Open Ended Module:		12	
	1	Practical problems based on module I to IV using statistical software.		
	Sections from References:			

Books and References:

Altman, D G. (2006): Practical Statistics for Medical Research, London: Chapman and Hall.

Cox, D.R. and Oakes, D. (1984): Analysis of Survival Data, Chapman and Hall.

Daniel, W.W.(2006): Biostatistics: A Foundation for Analysis in the Health sciences, John Wiley & sons. Inc.

Dunn, G. and Everitt B. (1995): Clinical Biostatistics: An Introduction to Evidence-based Medicine. Edward Arnold.

Friedman, L.M., Furburg, C. and DeMets, D.L. (1998): Fundamentals of Clinical Trials, Springer Verlag.

Gross, A. J. and Clark V.A. (1975): Survival Distribution; Reliability Applications in Biomedical Sciences, John Wiley & Sons.